



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df020-2017-ios
Operator:	Floris
Analysis Started:	2026-06-18 15:11:22
Analysis Completed:	2026-06-18 15:13:59
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df020\2017\dfdof_output\ft-df020-2017-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df020_external.001	drone_sd	physical	14.8 GB	2026-06-18 15:12:22
ios_logical.zip	controller_ios	logical	666.2 MB	2026-06-18 15:12:24

Evidence Source Path

C:\Users\Floris\Documents\ddji_drones\test_mavic_pro\df020\2017\ios

Cryptographic Hashes

df020_external.001		drone_sd	physical
SHA-256:	46de2036acf691e5c1cbeceb784993e236d61cf7487d6f8cd4057e491a843846		
SHA-1:	d9334ed9dc418533429bb281e20b6e4261067704		

ios_logical.zip		controller_ios	logical
SHA-256:	342f0a92ce0b2f20acc36cc50543fde75c16e1b7456c73431f2553e97406e67d		
SHA-1:	3d507bd020bd35873bb1c06d5476a37a20edc656		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - drone flight storage]: source evidence not found
4. [p4 - controller android]: source evidence not found
5. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	1	1	[OK]
exiftool	12	9	[~] Partial
fls	1	1	[OK]
icat	8	8	[OK]
mmls	1	1	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 1
Tools Succeeded	6 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	08RDE5F001007H	[K] Controller-only	2
Drone Model	MavicPro	[K] Controller-only	2
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	1
DJI Application Version	4.1.5	[K] Controller-only	2
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	13	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-08-29T18:14:14Z	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	medium
GPS Overlap:	451.7 s
Median GPS Distance:	9.67 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	400b92def473e678b41f1c11a1894245476c5ec5e163a52dcd7cf9b341738219
Flight Record Name:	DJIFlightRecord_2017-08-29_12-21-48_.csv
Start:	2017-08-29T18:21:48.323000+00:00
End:	2017-08-29T18:29:19.987000+00:00
Duration:	451.7 s
Drone Log SHA-256:	5c1e7d4d801ac44bfb288a216a965c3256c9bf5dcdac140ef74dda657004856d
Drone Log Name:	2017-08-29_12-20-53_FLY003.csv
Start:	2017-08-29T18:20:39.904000+00:00
End:	2017-08-29T18:29:23.688000+00:00
Duration:	523.8 s

Flight Metrics

Peak Height:	54.731241407598 m
Distance Travelled:	2072.91 m
Video Recording:	Yes
Photo Capture:	Yes
Photos Taken:	2
Recording Duration:	429.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. Atti
2. GPS_Atti
3. ASST_TAKEOFF
4. AutoTakeoff
5. EngineStart
6. Joystick
7. NaviSubMode_Pointing
8. GoHome
9. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 18:20:51	ctrl_ios:dl	Log started	MavicPro
2017-08-29 18:20:51	ctrl_ios:dl	Motor turned off	off
2017-08-29 18:21:46	ctrl_ios:dl	Motor turned on	on
2017-08-29 18:21:48	ctrl_ios:fr (v4.1.5)	Log started	SN: 08RDE5F001007H; App: 4.1.5; Mavic Pro
2017-08-29 18:21:48	ctrl_ios:fr (v4.1.5)	Motor turned on	on
2017-08-29 18:21:49	ctrl_ios:fr (v4.1.5)	Record mode changed	No
2017-08-29 18:21:53	ctrl_ios:fr (v4.1.5)	Record mode changed	Start
2017-08-29 18:21:54	ctrl_ios:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 18:26:17	ctrl_ios:dl	Reached peak height	54.7 m
2017-08-29 18:26:18	ctrl_ios:fr (v4.1.5)	Reached peak height	56.8 m
2017-08-29 18:27:05	ctrl_ios:fr (v4.1.5)	Record mode changed	Stop
2017-08-29 18:27:06	ctrl_ios:fr (v4.1.5)	Record mode changed	No

Timestamp (UTC)	Source	Event	Details
2017-08-29 18:27:22	ctrl_ios:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 18:29:18	ctrl_ios:dl	Motor turned off	off
2017-08-29 18:29:19	ctrl_ios:fr (v4.1.5)	Log ended	Dist home: 1.0 m; Height: 2.0 m; Rec: 429.0 s
2017-08-29 18:29:24	ctrl_ios:dl	Log ended	
2017-08-29 18:27:19	ctrl_ios:fr (v4.1.5)	Photo mode changed	Single

In-Flight Warnings

Timestamp (UTC)	Source	Warning Message
2017-08-29 18:23:12	ctrl_ios:fr (v4.1.5)	Warning:In Flight.workingcompasscountersmagnetic-field interference.please switch to atti mode if craft behave abnormally
2017-08-29 18:26:01	ctrl_ios:fr (v4.1.5)	Warning:In Flight.workingcompasscountersmagnetic-field interference.please switch to atti mode if craft behave abnormally
2017-08-29 18:26:05	ctrl_ios:fr (v4.1.5)	Warning:Compass Redundancy Switch

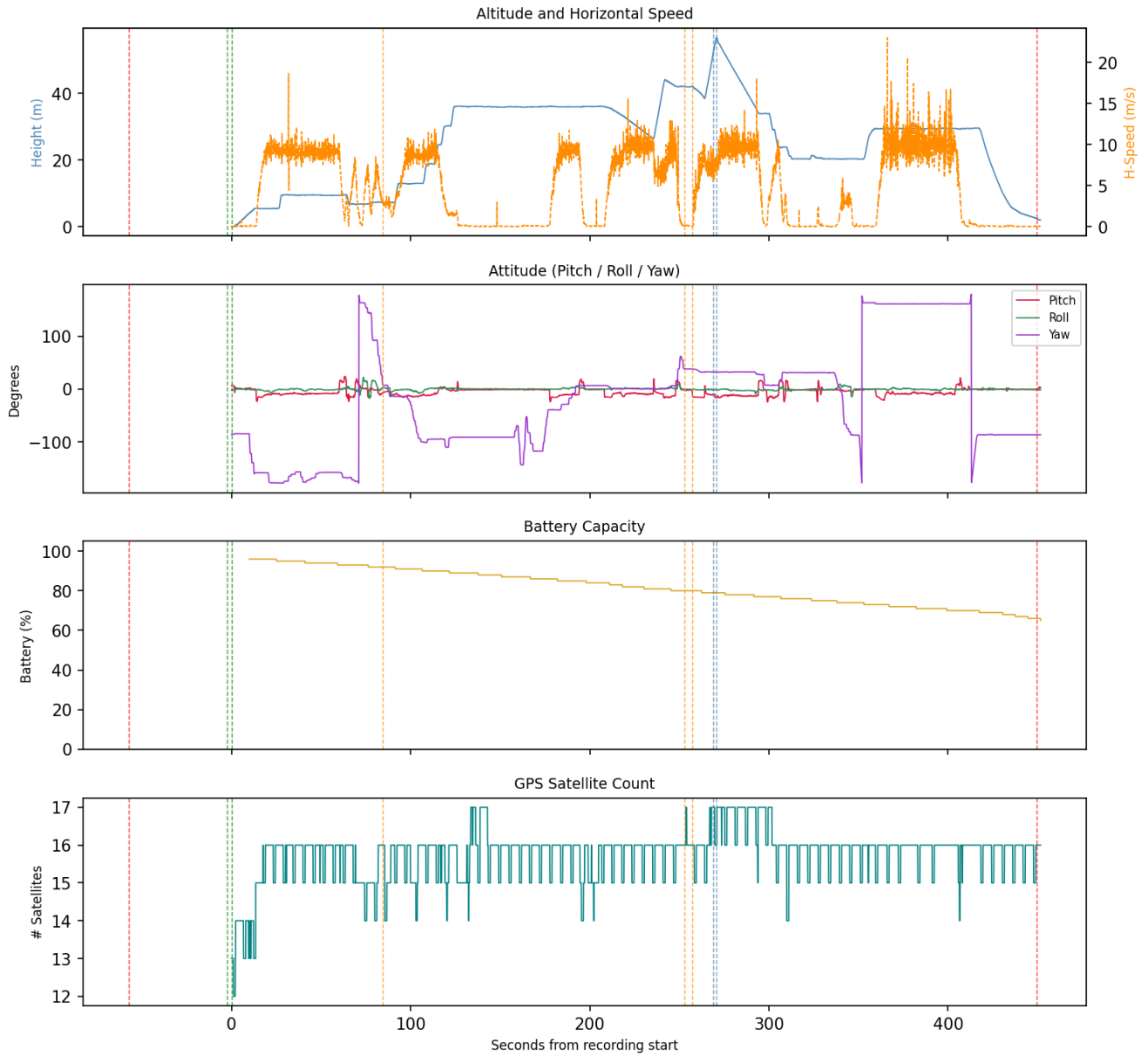
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:68eb24a05eae9a71ef79f0e865dd4b8f928db737655e282b0958b2462c972b02	
drone_sd:videos	GPS within flight bounding box (lat 39.9649, lon -106.2182)
p5:c28ad8959bcacd1b61c6d5425e89bdfe480d643e315a65ddbd2b8c942229c67	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2164)
p5:0efec08bdc3058e73f76114f653680f509c8f6f9e8e380ca3057bf7453c8bffd	

Flight Record - Sensor Telemetry

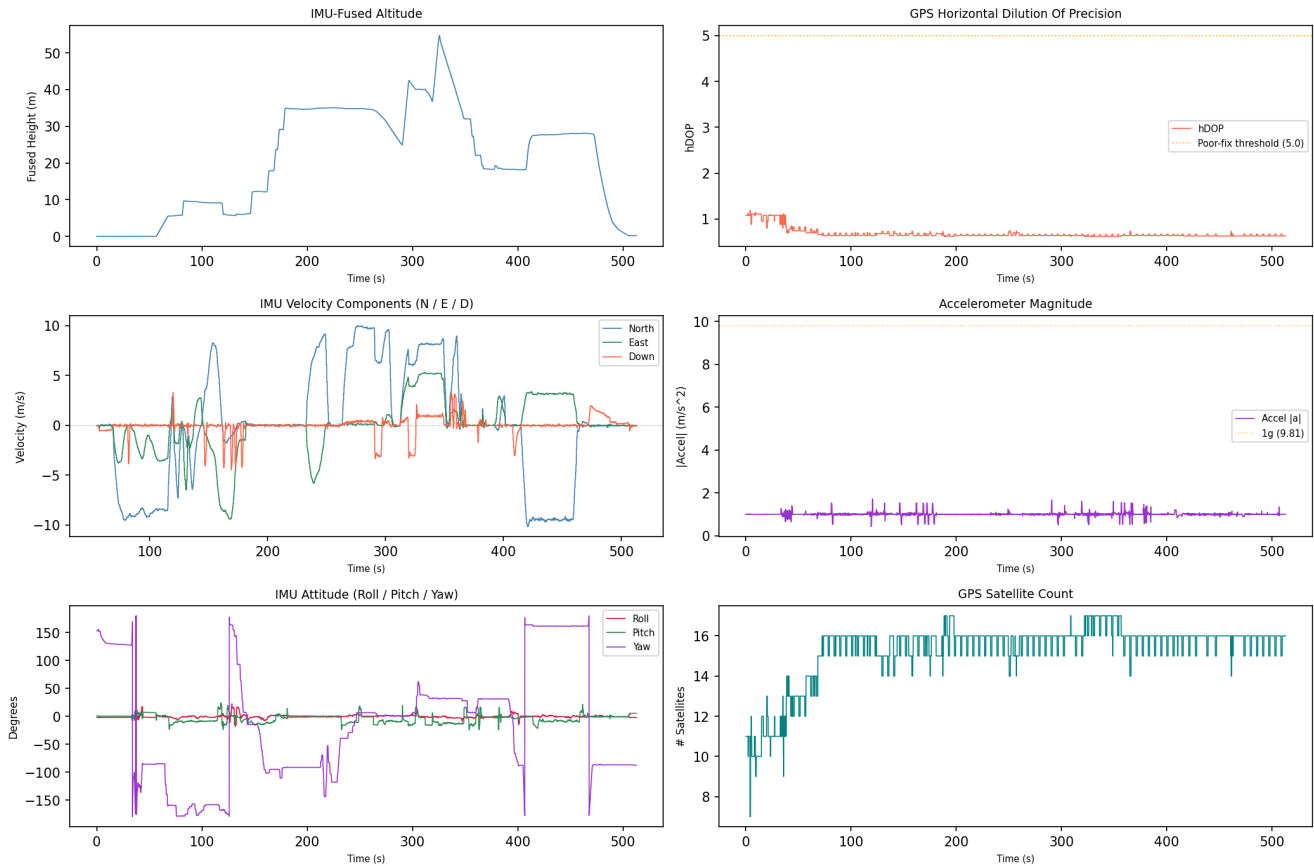
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

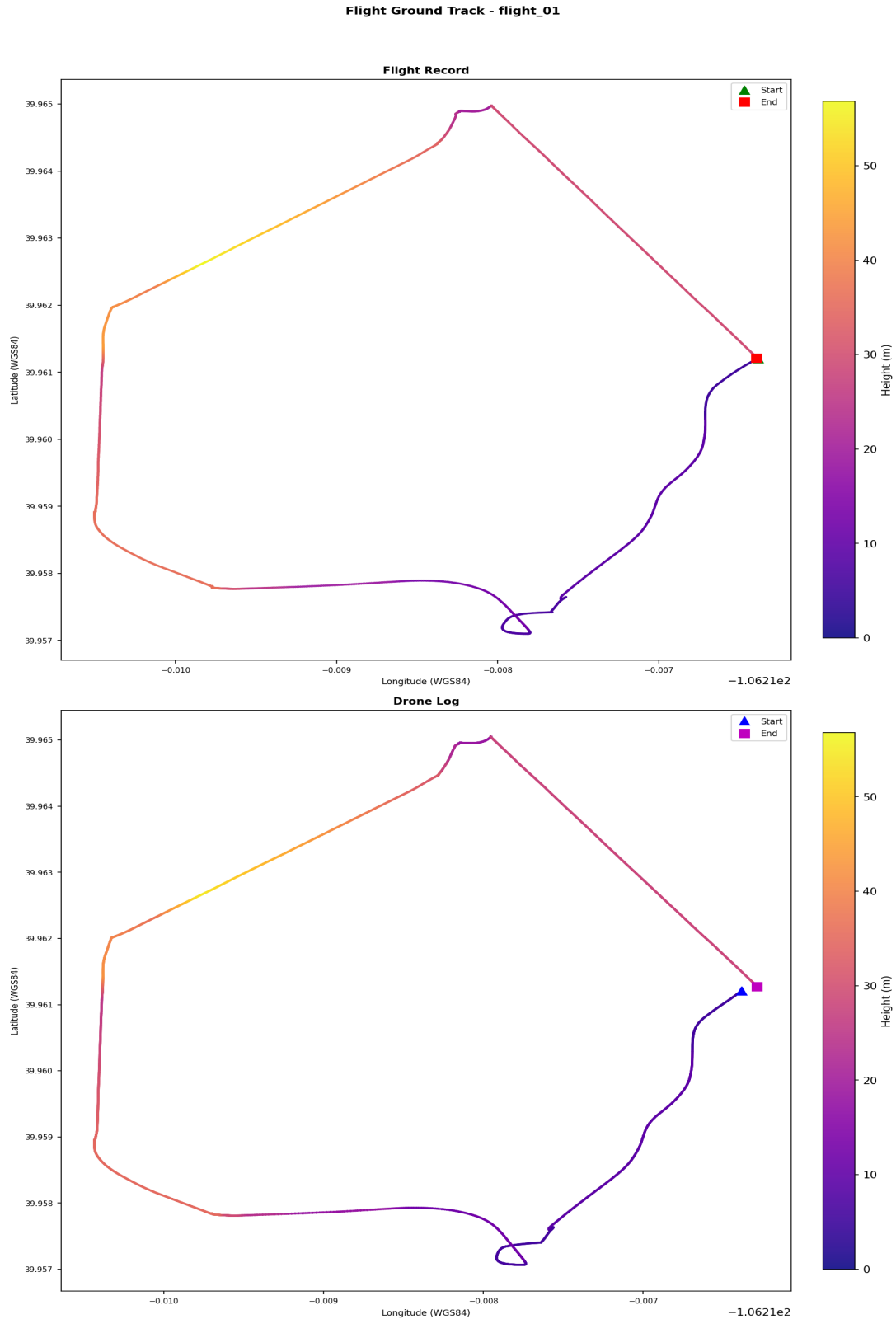
Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2017-08-29_12-20-53_FLY003.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	1	6	1	4	1	2	2
drone_sd	0	0	0	0	0	5	3

Data Quality Notes per Category

Databases (6 files):

- 2 database(s) empty (controller_ios)

Drone Logs (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Flight Logs (4 files):

- 1 file(s) unreadable (binary)
- formats: 2 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (7 files):

- 2 file(s) contain(s) EXIF GPS coordinates
- 7 file(s) missing (normalisable) EXIF timestamp
- 5 file(s) missing EXIF GPS coordinates

Videos (5 files):

- 3 file(s) contain(s) EXIF GPS coordinates
- 5 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 1 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 1 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 2 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - drone flight storage]: source evidence not found
4. [p4 - controller android]: source evidence not found
5. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

statistics.db

- database_empty: all tables empty

Drone Logs

norm_2017-08-29_12-20-53_FLY003.csv

- missing_gps: 3 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 30 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl:f:D (+25 more)

Flight Logs

2017-08-29_12_21_48-08RDE5F001007H.dat

- format: json_array_log | entries: 13

upgradeLog_0.txt

- format: bracketed_log | entries: 32

upgradeLog_1.txt

- format: bracketed_log | entries: 194

Flight Records

norm_DJIFlightRecord_2017-08-29_12-21-48.csv

- altitude_spike: 1 row(s) affected
- contains_no_value: 39 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+34 more)
- contains_constant_value: 157 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType, CAMERA_INFO.firmUpgradeState, CAMERA_INFO.hotState (+152 more)

Images

2017_08_29_12_21_53.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_12_27_22.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2017_08_29_12_21_53.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_08_29_12_27_22.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no firmware version found - Device and account information.
2. Note: no last known latitude (find aircraft) found - Device and account information.
3. Note: no last known longitude (find aircraft) found - Device and account information.
4. Note: no last known altitude (find aircraft) found - Device and account information.
5. Note: no last known heading (find aircraft) found - Device and account information.
6. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
7. Photo(s) were recorded during the flight. Copies may reside on both the controller and the drone SD card - both sources should be examined.
8. 4 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	23c29dca2106e224e167618a34a0186e43b01d6fd3fb8f532ccf7d688a11e455			
SHA-1:	fc402d5f165a725d87dd3c2622dfcf4404050710			

airmap_flysafepplaces.db		Databases	controller_ios	extract_logical
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c			
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	c3635c4495369db4324667ff800fa21938ab44d19853d7cf5c462d5795a4eebe			
SHA-1:	5fbc7f7d19f443b48975f45ee1a0ef6abb3f12f3			

nfz_flysafepplaces.db		Databases	controller_ios	extract_logical
SHA-256:	abcc5102acafb4c59ea39ad317304f6100951de9d71bfe51b73ef32be273ada7			
SHA-1:	78d1709ae8f46d404297904d0fb0f258c873517e			

statistics.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdfa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

tencent_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	fa412c96b97c0dd7a7567ad0203a11c28b91bb2ee474bbe97a56c1937f893522			
SHA-1:	0a9a7917e01d2f5d2a05d76f3c799cf8667d3ac8			

tencent_wxo_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	273d9a72c1bc54a860f56657335dcea25fc1cc69e3f6b040cca32f8228d5ee5d			
SHA-1:	94d9e466fe36375fe26dc22280a0ba94d4ad8d64			

2017-08-29_12-20-53_FLY003.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	2e1ddda2cf92a52b72060c37a39e6d14f8e3c61607d73471b1fef87ea9046e57			
SHA-1:	871991b296c31308c34ed934766c5d7d2b8ab991			

2017-08-29_12_21_48-08RDE5F001007H.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	f7700e9079729efb1e659782d2a8d7ced3f62635741b5ae12d5c7027bc1bc907			
SHA-1:	fc73a3916e9a9462e761f81e23fa5685c77b7962			

app_log.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	a8518b013f7fc839d4d0794bc700ab8350d445bff983b1ab8d0922b82d097956			
SHA-1:	fe08dc2bc871ed7a5a73705ac338b681a1a8f8f8			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	7f8b4116ec7ab2313bf24b08b138eb70733818582713f49cc493359935804585			
SHA-1:	d752db20ebf24edad4d8de4c668bd3b2e777cea0			

upgradeLog_1.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	ce234e3d610c3f47fdc32e45f0829130460177ae40efe8f612f732f0582b479b			
SHA-1:	9dff80b25315de80dfcd743859039d484551be76			

DJIFlightRecord_2017-08-29_12-21-48_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	50e7fcdef6db9490af4b058330ed0c03d2f3eccc80168fed4b263b431aef5c5a		
SHA-1:	7d4b9da83c868e858ec8f5f3299ed8994a2ffc20		

2017_08_29_12_21_53.thumbnail	Images	controller_ios	extract_logical
SHA-256:	24d2984d08a2744a8bcd18da464e9512130777e41305368e48abbfe1984e2c3d		
SHA-1:	ebad66f800b51f9f5f8d46072058910400b15036		

2017_08_29_12_27_22.thumbnail	Images	controller_ios	extract_logical
SHA-256:	8601de44d4fc6e9a2eff3159b2ec004e715d46369142242f7c9a3112d4481ca8		
SHA-1:	89c1cae9cb974fd6c72e8988248c8168408b1eaf		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	4e23f027a9017ebb6a1e2714423dff8816985fc9388b08e48e206e6f4190bd7d		
SHA-1:	c331731ce9baafc7e0d8e98ee1b5eed3017ba821		

DJI_0002.JPG	Images	drone_sd	extract_physical
SHA-256:	d0f284faea570872fd451c0906bd1985e58e152cabe271f7ce80e35fe1b5d142		
SHA-1:	b3753796ce2c07730405516cbb8acd57f2580f13		

DJI_0003.JPG	Images	drone_sd	extract_physical
SHA-256:	da2f39ae8ae220ea0567e60e871da87b1ef7c2f66bf1aefb66d86eac3d1a0dd2		
SHA-1:	c5b47d8f4288258a444aa33e90a8523624be34dc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	d6de6e1becc1aac8bd084968d48e43b958a1d91e9c43e7643897cda189800a7f		
SHA-1:	7e21ddd036c409dde759460dc4e2740c7b96da1e		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b254d105f743634984b958b82ee4730f71b60783f9511a6585ca9da5e3410704		
SHA-1:	5c49385e227e85aa668aafd0cdfa57b7e51d409f		

2017_08_29_12_21_53.mp4	Videos	controller_ios	extract_logical
SHA-256:	24b23cb611160ad6bd0596566718fba6009ebad759222ab989a2b72623d17b3d		
SHA-1:	078e52893f74f7176fdf6c3565c531c549bdfb35		

2017_08_29_12_27_22.mp4	Videos	controller_ios	extract_logical
SHA-256:	77bf272e93b9ab1c42575451d8d605729371e1577ccbd20aa6aac85b01cbcd31		
SHA-1:	731c3d5567ba6e1013fed805df9ba20a8ecdf820		

DJI_0001.MOV	Videos	drone_sd	extract_physical
SHA-256:	68eb24a05eae9a71ef79f0e865dd4b8f928db737655e282b0958b2462c972b02		
SHA-1:	25040bb7c7a9d18e8afa5683d627971348241a57		

DJI_0004.MOV	Videos	drone_sd	extract_physical
SHA-256:	c28ad8959bcacd1b61c6d5425e89bdfc480d643e315a65ddbd2b8c9422229c67		
SHA-1:	69de46ec683056c48e84ada8104d9db5893f20a9		

DJI_0005.MOV	Videos	drone_sd	extract_physical
SHA-256:	0efec08bdc3058e73f76114f653680f509c8f6f9e8e380ca3057bf7453c8bfffd		
SHA-1:	c8cad52303de51cb075924924f82e2538aab6239		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	6	0	0
Drone Logs	1	2	1
Flight Logs	4	0	0
Flight Records	1	1	1
Images	2	2	2
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	5	5	0
Videos	3	3	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 15:11:22 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df020_external.001

2026-06-18 15:11:34 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df020_external.001

2026-06-18 15:12:31 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdo_output\ft-df020-2017-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-18 15:12:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8197
out: DJI_0001.MOV

2026-06-18 15:12:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8198
out: DJI_0002.JPG

2026-06-18 15:12:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8199
out: DJI_0003.JPG

2026-06-18 15:12:50 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8200
out: DJI_0004.MOV

2026-06-18 15:13:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 8201
out: DJI_0005.MOV

2026-06-18 15:13:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9222
out: DJI_0001.THM

2026-06-18 15:13:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9223
out: DJI_0004.THM

2026-06-18 15:13:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df020_external.001 9224
out: DJI_0005.THM

2026-06-18 15:13:46 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2017-08-29_12-20-53_FLY003.DAT
out: 2017-08-29_12-20-53_FLY003.csv, 2017-08-29_12-20-53_FLY003.kml

2026-06-18 15:13:48 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-08-29__12-21-48_.txt DJIFlightRecord_2017-08-29__12-21-48_.csv
out: DJIFlightRecord_2017-08-29__12-21-48_.csv

2026-06-18 15:13:48 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_21_53.thumbnail
out: 2017_08_29_12_21_53_exif.json

2026-06-18 15:13:49 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_27_22.thumbnail
out: 2017_08_29_12_27_22_exif.json

2026-06-18 15:13:49 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_21_53.mp4
out: 2017_08_29_12_21_53_exif.json

2026-06-18 15:13:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_27_22.mp4
out: 2017_08_29_12_27_22_exif.json

2026-06-18 15:13:50 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-18 15:13:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-18 15:13:51 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG
out: DJI_0003_exif.json

2026-06-18 15:13:51 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-18 15:13:51 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-18 15:13:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV
out: DJI_0001_exif.json

2026-06-18 15:13:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MOV
out: DJI_0004_exif.json

2026-06-18 15:13:53 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV
out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.